⊗ Resettable and settable flip-flops may also have enable input and may be grounded into N-bit registers.

| RESET | CLK | D | Q(next) |
|---|---|---|---|
| 0 | ↓ | 0 | 0 |
| 0 | ↑ | 1 | 1 |
| 1 | ↓ | X | 0 |

## Problems with sequential Logic:-

1) Loops are unstable
2) Memory elements can suffer from race conditions.
3) Output .. depends upon input sequence.

Solution:- Inserting a flip flop into the cyclic loop. Race conditions are eliminated if the clock time period is long enough.

Race Conditions:- In Asynchronous circuits are famous for having race conditions where the behaviour of the circuit depends on which of two paths through logic gates is fastes.

Cycle Fo

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- Sequential feedback
- These fo conditions
- Analysing lead to

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- Every logic.
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